

DAY 12

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Task:

1. To the existing books table, add additional column called price and add at least ten entries with some price values.

1.1 Create a post request 'GetBooksOrderedByPrice()' with a boolean parameter

1.1.1 when parameter is true, then sort by price ascending order

1.1.2 when parameter is false, then sort by price descending order

1.2 Create a get request 'GetTop5Books()' which will return top 5 costliest books

1.3 Create a get request 'GetCostDetails()' which will return the total price and average price of all books.

Code:

Book.cs

```
using System.ComponentModel;
using System.ComponentModel.DataAnnotations;

namespace Day12Api.Models
{
    public class Book
    {
        public int Id { get; set; }

        [DisplayName("Book Name")]
        [StringLength(60, MinimumLength = 6, ErrorMessage = "Must be atleast 6 characters in length")]
        public string Title { get; set; }
        public string Description { get; set; }

        [Range(0.0, 5.0)]
        public float Rating { get; set; }
        public int Price { get; set; }
    }
}
```

AppDbContext.cs

```
using Day12Api.Models;
using Microsoft.EntityFrameworkCore;

namespace Day12Api.Context
{
    public class AppDbContext : DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options) : base(options)
        {
        }

        public DbSet<Book> Books { get; set; }

        public DbSet<User> Users { get; set; }
    }
}
```

BookController.cs

```
using Day12Api.Context;
using Day12Api.Models;
using Microsoft.AspNetCore.Mvc;

namespace Day12Api.Controllers
{
    [ApiController]
    [Route("[controller]")]
    public class BookController : Controller
    {
        AppDbContext _appDbContext;

        public BookController(AppDbContext appDbContext)
        {
            _appDbContext = appDbContext;
        }

        [HttpGet("GetAllBooks")]
        public IEnumerable<Book> GetAllBooks()
        {
            var all_books = _appDbContext.Books.ToList();
            return all_books;
        }

        [HttpPost("AddBook")]
        public IActionResult AddBook(Book book)
        {
        }
    }
}
```

```

        _appDbContext.Books.Add(book);
        _appDbContext.SaveChanges();
        return Ok("Added book to the store");
    }

    [HttpDelete("RemoveBook")]
    public IActionResult DeleteBook(int id)
    {
        if (id < 0)
            return BadRequest("only positive numbers accepted.");
        var book_to_delete = _appDbContext.Books.FirstOrDefault(b => b.Id == id);
        if (book_to_delete == null)
            return BadRequest("Book not found");
        _appDbContext.Books.Remove(book_to_delete);
        _appDbContext.SaveChanges();
        return Ok("Deleted");
    }

    [HttpPost("GetBooksOrderedByPrice")]
    public IActionResult GetBooksOrderedByPrice(bool ascending)
    {
        var books = ascending
            ? _appDbContext.Books.OrderBy(b => b.Price).ToList()
            : _appDbContext.Books.OrderByDescending(b => b.Price).ToList();

        return Ok(books);
    }

    [HttpGet("GetTop5Books")]
    public IActionResult GetTop5Books()
    {
        var books = _appDbContext.Books
            .OrderByDescending(b => b.Price)
            .Take(5)
            .ToList();

        return Ok(books);
    }

    [HttpGet("GetCostDetails")]
    public IActionResult GetCostDetails()
    {
        var totalPrice = _appDbContext.Books.Sum(b => b.Price);
        var avgPrice = _appDbContext.Books.Average(b => b.Price);

        return Ok(new { TotalPrice = totalPrice, AveragePrice = avgPrice });
    }
}

```

Output:

When true, displays the book in ascending order by price

Name	Description
ascending boolean (query)	true

Execute

```
[
  {
    "id": 11,
    "title": "Animal Farm",
    "description": "about animals overthrowing humans, symbolizing political tyranny",
    "rating": 3.5,
    "price": 320
  },
  {
    "id": 12,
    "title": "Life of Pi",
    "description": "A boy survives at sea with a Bengal tiger in a lifeboat",
    "rating": 3.9,
    "price": 370
  },
  {
    "id": 13,
    "title": "Frankenstein",
    "description": "A scientist creates life, only to be haunted by his monstrous creation",
    "rating": 4,
    "price": 410
  },
  {
    "id": 14,
    "title": "Dracula",
    "description": "horror of Count Dracula's hunt for blood in England",
    "rating": 3.7,
    "price": 430
  }
]
```

```
},
{
  "id": 8,
  "title": "The Book Thief",
  "description": "about a girl who steals books",
  "rating": 4.3,
  "price": 450
},
{
  "id": 7,
  "title": "Little Women",
  "description": "story of four sisters navigating love, loss, and womanhood",
  "rating": 4.6,
  "price": 500
},
{
  "id": 5,
  "title": "The Kite Runner",
  "description": "story of friendship, betrayal, and redemption in Afghanistan",
  "rating": 4.5,
  "price": 560
},
{
  "id": 10,
  "title": "Jane Eyre",
  "description": "about independence, morality, and true love",
  "rating": 4.7,
  "price": 580
},
{
  "id": 9,
  "title": "The Alchemist",
  "description": "about following one's dreams and destiny",
  "rating": 4.9,
  "price": 630
},
{
  "id": 6,
  "title": "The Hobbit",
  "description": "fantasy adventure of Bilbo Baggins and a dragon's treasure",
  "rating": 4.7,
  "price": 700
}
]
```

When false, displays the book in descending order by price

Name	Description
ascending boolean (query)	false

Execute

```
[
  {
    "id": 6,
    "title": "The Hobbit",
    "description": "fantasy adventure of Bilbo Baggins and a dragon's treasure",
    "rating": 4.7,
    "price": 700
  },
  {
    "id": 9,
    "title": "The Alchemist",
    "description": "about following one's dreams and destiny",
    "rating": 4.9,
    "price": 630
  },
  {
    "id": 10,
    "title": "Jane Eyre",
    "description": "about independence, morality, and true love",
    "rating": 4.7,
    "price": 580
  },
  {
    "id": 5,
    "title": "The Kite Runner",
    "description": "story of friendship, betrayal, and redemption in Afghanistan",
    "rating": 4.5,
    "price": 560
  },
]
```

```
{
  "id": 7,
  "title": "Little Women",
  "description": "story of four sisters navigating love, loss, and womanhood",
  "rating": 4.6,
  "price": 500
},
{
  "id": 8,
  "title": "The Book Thief",
  "description": "about a girl who steals books",
  "rating": 4.3,
  "price": 450
},
{
  "id": 14,
  "title": "Dracula",
  "description": "horror of Count Dracula's hunt for blood in England",
  "rating": 3.7,
  "price": 430
},
{
  "id": 13,
  "title": "Frankenstein",
  "description": "A scientist creates life, only to be haunted by his monstrous creation",
  "rating": 4,
  "price": 410
},
{
  "id": 12,
  "title": "Life of Pi",
  "description": "A boy survives at sea with a Bengal tiger in a lifeboat",
  "rating": 3.9,
  "price": 370
},
{
  "id": 11,
  "title": "Animal Farm",
  "description": "about animals overthrowing humans, symbolizing political tyranny",
  "rating": 3.5,
  "price": 320
}
]
```

The top 5 costliest books

```
[
  {
    "id": 6,
    "title": "The Hobbit",
    "description": "fantasy adventure of Bilbo Baggins and a dragon's treasure",
    "rating": 4.7,
    "price": 700
  },
  {
    "id": 9,
    "title": "The Alchemist",
    "description": "about following one's dreams and destiny",
    "rating": 4.9,
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  },
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    "id": 10,
    "title": "Jane Eyre",
    "description": "about independence, morality, and true love",
    "rating": 4.7,
    "price": 580
  },
  {
    "id": 5,
    "title": "The Kite Runner",
    "description": "story of friendship, betrayal, and redemption in Afghanistan",
    "rating": 4.5,
    "price": 560
  },
  {
    "id": 7,
    "title": "Little Women",
    "description": "story of four sisters navigating love, loss, and womanhood",
    "rating": 4.6,
    "price": 500
  }
]
```


The total and average price of all books

```
{
  "totalPrice": 4950,
  "averagePrice": 495
}
```

List of Books - Database

Results		Messages			
	Id	Title	Description	Rating	Price
1	5	The Kite Runner	story of friendship, betrayal, and redemption in Afg...	4.5	560
2	6	The Hobbit	fantasy adventure of Bilbo Baggins and a dragon'...	4.7	700
3	7	Little Women	story of four sisters navigating love, loss, and wom...	4.6	500
4	8	The Book Thief	about a girl who steals books	4.3	450
5	9	The Alchemist	about following one's dreams and destiny	4.9	630
6	10	Jane Eyre	about independence, morality, and true love	4.7	580
7	11	Animal Farm	about animals overthrowing humans, symbolizing p...	3.5	320
8	12	Life of Pi	A boy survives at sea with a Bengal tiger in a lifeb...	3.9	370
9	13	Frankenstein	A scientist creates life, only to be haunted by his m...	4	410
10	14	Dracula	horror of Count Dracula's hunt for blood in England	3.7	430